Arista Access Node WFD-ACC-31 Hardware/Environmental Fault Failure Resolution SOP

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8060

Description

The occurrence of hardware/environmental faults on the Arista Access Node WFD-ACC-31, manifesting as ASIC memory parity errors, line card crashes, and power supply redundancy loss, is a critical failure domain that warrants immediate attention. This failure domain is specific to the Arista Access Node equipment due to its reliance on the EOS (Extensible Operating System) software framework, which leverages the ASIC (Application-Specific Integrated Circuit) for forwarding decisions. The ASIC memory parity errors are typically indicative of a hardware failure, while line card crashes and power supply redundancy loss can be caused by environmental factors such as overheating, power surges, or electrical noise. The Arista Access Node's architecture, which employs a distributed control plane and a centralized data plane, makes it susceptible to hardware faults that can propagate across the control and data planes. Furthermore, the Arista Access Node's reliance on the EOS software framework, which is designed to provide a high level of scalability and flexibility, can sometimes mask hardware faults, making it essential to employ a combination of diagnostic commands and physical checks to identify and resolve the root cause of the failure.

Pre-requisite Checks

- Physical Checks: Verify that the power supplies are functioning correctly and that there are no signs of overheating or electrical noise in the vicinity of the Arista Access Node.

- Environmental Checks: Check the temperature and humidity levels in the Whitefield POP to ensure that they are within the recommended operating ranges for the Arista Access Node.

- Remote Hands: Dispatch remote hands to the Whitefield POP to perform further diagnostic checks and to provide physical access to the Arista Access Node if required.

- Redundant Link Status: Verify that the redundant links are functioning correctly and that there are no issues with the link aggregation protocol.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal status, Expected Output: System status and version information., Description: This command provides an overview of the system status, including the ASIC version and the number of forwarding engines.

Command: show system internal forwarding-table, Expected Output: Forwarding table information, including the ASIC memory usage., Description: This command provides detailed information about the forwarding table, including the ASIC memory usage and the number of entries.

Command: show system internal power-supply, Expected Output: Power supply status and redundancy information., Description: This command provides information about the power supply status and the redundancy level.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace system internal asic , Description: This step involves replacing the faulty ASIC with a new one, which can be achieved using the replace system internal asic command. The asic_slot_number parameter specifies the slot number of the faulty ASIC.

Step 2: Step: Replace the faulty line card, Command: replace system internal line-card , Description: This step involves replacing the faulty line card with a new one, which can be achieved using the replace system internal line-card command. The line_card_slot_number parameter specifies the slot number of the faulty line card.

Step 3: Step: Replace the power supply unit, Command: replace system internal power-supply , Description: This step involves replacing the faulty power supply unit with a new one, which can be achieved using the replace system internal power-supply command. The power_supply_slot_number parameter specifies the slot number of the faulty power supply unit.

Post-Check Verification


```
Command: show system internal status, Expected Output: System status and version information., Description: This command provides an overview of the system status, including the ASIC version and the number of forwarding engines.

Command: show system internal forwarding-table, Expected Output: Forwarding table information, including the ASIC memory usage., Description: This command provides detailed information about the forwarding table, including the ASIC memory usage and the number of entries.

Command: show system internal power-supply, Expected Output: Power supply status and redundancy information., Description: This command provides information about the power supply status and the redundancy level.
```


Roll-back Procedures

- Step: Restore the original ASIC, Command: restore system internal asic , Description: This step involves restoring the original ASIC from the backup, which can be achieved using the restore system internal asic command. The asic_slot_number parameter specifies the slot number of the original ASIC.

- Step: Restore the original line card, Command: restore system internal line-card , Description: This step involves restoring the original line card from the backup, which can be achieved using the restore system internal line-card command. The line_card_slot_number parameter specifies the slot number of the original line card.

- Step: Restore the original power supply unit, Command: restore system internal power-supply , Description: This step involves restoring the original power supply unit from the backup, which can be achieved using the restore system internal power-supply command. The power_supply_slot_number parameter specifies the slot number of the original power supply unit.